

Mechanical Design Considerations

The structural design must prioritize stiffness to prevent “kinematic singularity” or binding under load.

- **Cantilever vs. Yoke:** Design candidates often include a “Cantilever” style (supporting the leg from one side) to minimize the width between legs (Q-Factor), or a “Yoke” style (supporting from both sides) for maximum rigidity.
- **Load Paths:** The mechanism must effectively transfer the ground reaction forces from the leg into the torso chassis without significant deflection.

Actuation & Transmission Strategy

- **Torque Requirements:** The Roll and Pitch axes require massive torque (often exceeding 30–40 Nm peak) to recover from stumbles.
- **Actuator Candidates:** High-torque density solutions are essential. Designs often utilize Large-Stator Brushless DC (BLDC) motors (e.g., pancake style) to maximize torque generation.
- **Transmission Examples:** To convert motor speed to torque, transmission systems such as Single-Stage Planetary Gears (for compactness) or Cycloidal Drives (for shock load resistance) are frequently analyzed. A Quasi-Direct Drive (QDD) approach with low gearing ratios (e.g., 6:1 to 10:1) is preferred to maintain back-drivability.

Sensing & Feedback

- **Position:** High-resolution absolute encoding (14-bit or higher) is mandatory to ensure the leg angle is known precisely on startup without calibration movement.
- **Thermal:** Due to high current draw during balancing, integrated thermistors are necessary to monitor coil temperature and trigger thermal throttling algorithms.

3.3 The Knee Joint

Functional Objective: The Knee is the locomotive “spring” of the robot. Unlike industrial robotic knees which are designed for static positioning, a humanoid knee must

be a dynamic energy storage and dissipation unit. Its primary function is to manage impact energy during the heel-strike phase of walking and to provide explosive extension force during the push-off phase (jumping or running).

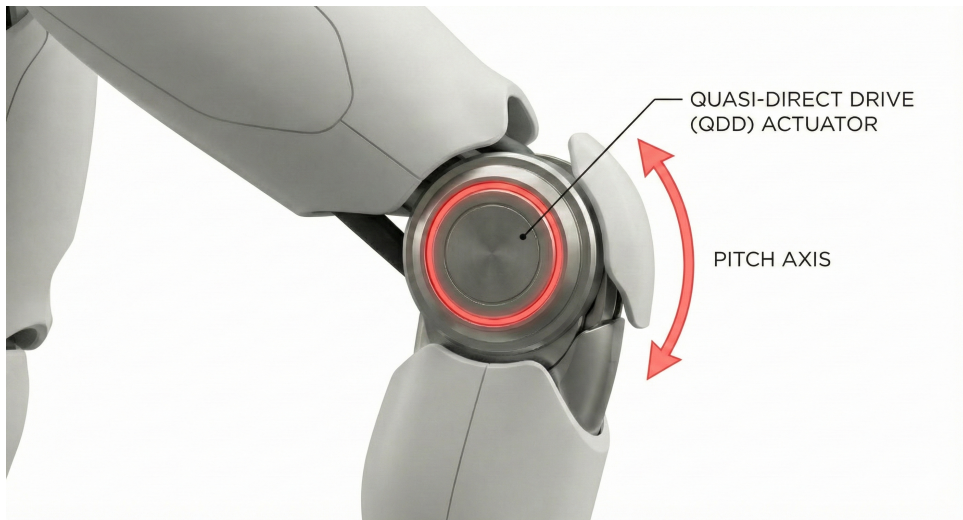


Figure 3.3: Knee joint architecture illustrating the single degree of freedom hinge mechanism and Quasi-Direct Drive actuation concept.

Kinematic Architecture (1-DOF)

- **Knee Pitch (Y-Axis):** A pure hinge motion. The range of motion typically spans from 0° (straight) to approximately 140° (deep squat) to allow for versatile maneuvers like standing up from a chair or recovering from a fall.

Mechanical Design Considerations

- **Impact Resistance:** This joint takes the brunt of the “Shock Load” when the foot hits the ground. The design must isolate the motor shaft from these radial and axial forces.
- **Bearing Architecture:** To protect the actuator, the output shaft is usually supported by robust bearing assemblies. Examples include Crossed-Roller Bearings for compact handling of multi-directional moments, or Dual Angular Contact Bearings spaced widely apart to resist bending forces.

Actuation & Transmission Strategy (The “QDD” Approach)

The Problem with High Gearing: Traditional high-ratio gearboxes ($>100:1$) act as mechanical brakes; shockwaves cannot travel backward through them, leading to gear

teeth failure during impact.

The Solution (Back-drivability): The design prioritizes Quasi-Direct Drive.

- **Motor Candidates:** Low-KV, high-torque BLDC motors (often used in heavy-lift drones) are ideal candidates due to their “torque-per-amp” efficiency.
- **Transmission Candidates:** A low-ratio Planetary Gearbox (e.g., 6:1 to 9:1) allows the joint to be “back-drivable.” If a human pushes the robot’s leg, the force spins the motor backward, generating current.

Sensing & Feedback

- **Proprioception:** By monitoring the current (Amps) and the precise rotor position, the controller can estimate the external torque applied to the knee.
- **Calibration:** An absolute magnetic encoder is typically placed on the output side (after the gearbox) or input side (motor shaft).

3.4 The Ankle Assembly

Functional Objective: The Ankle functions as the “Ground Adapter.” While the hip and knee generate large motions, the ankle makes micro-adjustments to keep the foot flat on uneven terrain. It is critical for the “Zero Moment Point” (ZMP) control strategy, where the robot manipulates the center of pressure to maintain balance.

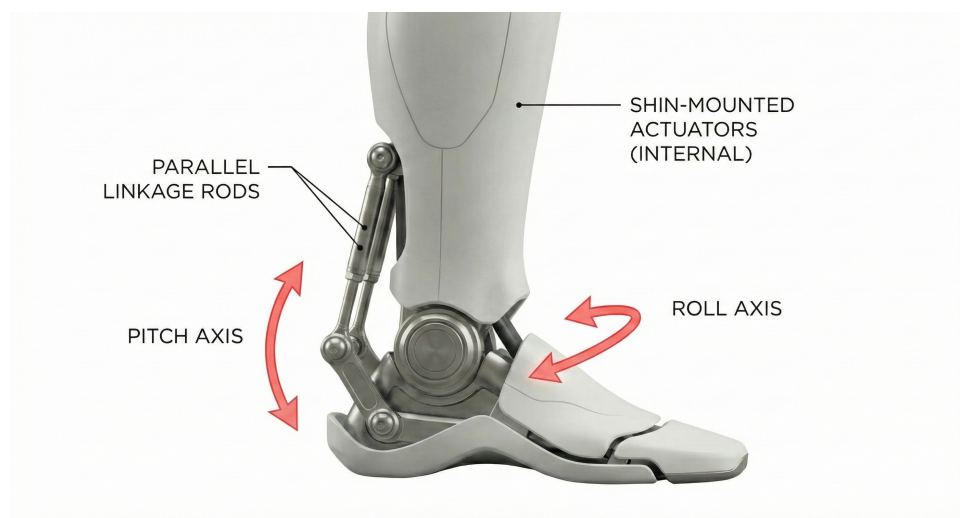


Figure 3.4: Ankle assembly with two degrees of freedom (Pitch and Roll) utilizing a parallel linkage and remote actuation strategy.

Kinematic Architecture (2-DOF)

- **Ankle Pitch (Sagittal Plane):** Controls dorsiflexion (toes up) and plantarflexion (toes down). This is used for propulsion and landing.
- **Ankle Roll (Coronal Plane):** Controls inversion/eversion. This allows the robot to walk on slanted surfaces or spread its legs for stability without riding on the edges of its feet.

Mechanical Design Considerations (Inertia Reduction)

The “Heavy Foot” Problem: Placing heavy motors directly inside the ankle joint acts like wearing concrete boots. It increases the leg’s moment of inertia, requiring massive power from the hip to swing the leg.

The Remote Actuation Solution: A preferred design strategy is to mount the motors proximally (higher up) in the Shin/Tibia structure.

- **Linkage Candidates:** Power is transmitted to the foot via rigid linkages. A Parallel Linkage mechanism using Carbon Fiber push-rods with spherical rod-ends (Heim joints) is a standard high-performance solution.

Actuation & Transmission Strategy

- **Parallel Kinematics:** By using two linear actuators (or rotary actuators with crank arms) moving in parallel, the system can achieve both Pitch and Roll. Moving actuators in unison creates Pitch; moving them differentially creates Roll.
- **Actuator Candidates:** Standard BLDC motors integrated into the shin, coupled with lead screws (linear) or crank-rockers (rotary).

Sensing & Feedback

- **Kinematic Solving:** Since the motion is parallel (non-linear), the software must compute the inverse kinematics of the linkage in real-time to determine the exact foot angle based on actuator positions.

3.5 The Leg Structure (Femur & Tibia)

Functional Objective: The Leg structures (Thigh/Femur and Shin/Tibia) are the skeletal chassis of the lower body. They are not merely connecting rods; they serve as structural housings for the actuators, electronics, and wiring. Their primary goal is to provide maximum stiffness to prevent bowing or twisting under load while adding minimal weight to the system.

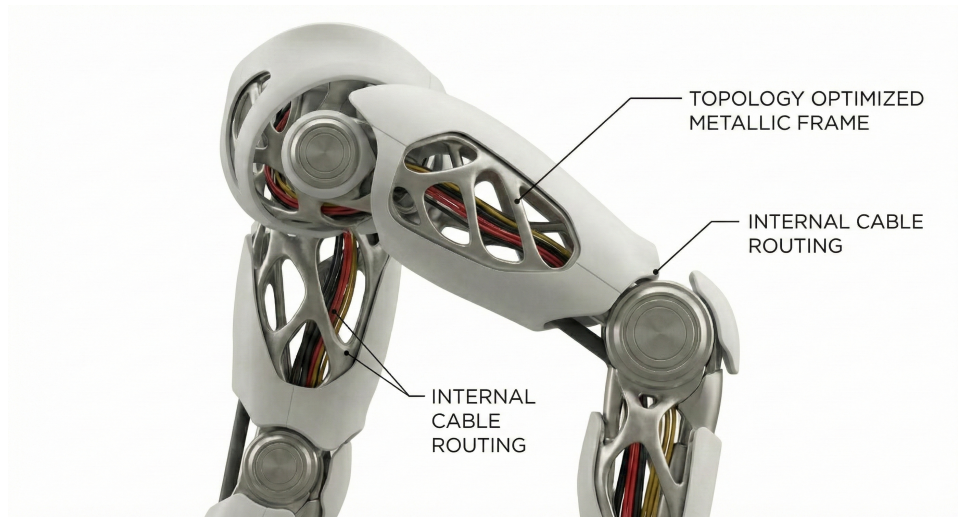


Figure 3.5: Leg structure showing femur and tibia with integrated actuator housing, cable routing, and hollow core design.

Mechanical Design Considerations

- **Topology Optimization:** Modern design rejects solid blocks of material. The geometry is often derived using generative design software, which simulates loads (compression from gravity, torsion from turning) and removes material from low-stress areas.
- **Torsional Stiffness:** The Femur must resist twisting when the robot turns. A closed-section profile (tube or box) is generally superior to an open section (plate).

Integration

- **Cable Routing:** The bones often feature internal channels or hollow cores. This allows the CAN-Bus and Power cables to run inside the leg, protecting them from snagging on the environment.

3.6 The Foot

Functional Objective: The Foot is the sole interface between the robot and the world. Its physics define the robot’s stability. It must provide high friction to prevent slipping, damping to absorb high-frequency vibration, and sensing to detect the Center of Pressure (CoP).

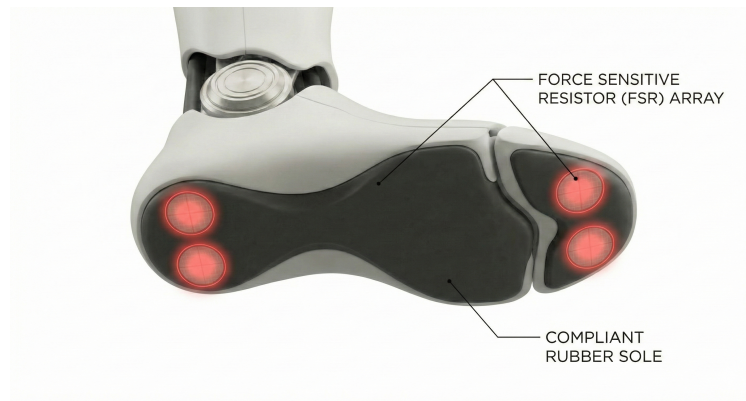


Figure 3.6: Foot design with support polygon, compliant sole, and integrated tactile sensing elements.

Mechanical Design Considerations

- **Geometry:** The foot is typically a flat plate to maximize the “Support Polygon” (the area in which the Center of Mass must stay to remain stable).
- **Passive Compliance (Damping):** A rigid metal foot hitting concrete creates dangerous shock spikes.

Sensing & Feedback (Tactile Interface)

Ground Contact Detection: The robot needs to know the exact millisecond the foot touches the floor to switch from “Position Control” (swing) to “Force Control” (stance).

Sensor Candidates:

- **Force Sensitive Resistors (FSRs):** Thin polymer films that change resistance when pressed. Placed at the 4 corners.
- **Load Cells:** Strain-gauge based sensors for high-precision weight measurement.
- **Tactile Skin:** Capacitive arrays that map the pressure distribution across the entire sole.

4. Systems & Core

4.1 Systems & Core Brief

The Systems & Core subsystem represents the physiological foundation of the robotic platform. While the upper and lower bodies define the robot’s physical capabilities, the Core defines its operational existence, transforming a collection of inert mechanisms into an autonomous agent. This section details the critical infrastructure required to bridge the gap between mechanical potential and intelligent action. It encompasses the high-density energy management systems (“Heart”), the heterogeneous computing architecture (“Brain”), the decision-making software stack (“Consciousness”), and the protective environmental interface (“Skin”). The primary engineering challenge of this subsystem is integration: the successful packaging of high-voltage power distribution networks alongside sensitive, low-latency AI processors within a confined, mobile chassis, all while effectively managing thermal loads and electromagnetic interference in real-time.

4.2 Heart & Arteries (Power & Distribution)

Functional Objective: The “Heart” of the robot is not merely a battery pack; it is a sophisticated Energy Management System (EMS) designed to handle extreme dynamic loads. During walking or jumping, the motors draw massive current spikes in milliseconds. The primary objective is to deliver this high instantaneous power without voltage sag, while simultaneously isolating the sensitive compute electronics from electrical noise. The “Arteries” represent the power and data distribution network, designed to route high-voltage power alongside low-voltage signals without interference.

Technical Design (Power Architecture)

- **High-Discharge Power Source:** The energy storage solution is selected for high power density (Watts/kg) rather than just energy density, ensuring explosive move-

ments without triggering over-current protection.

- **Isolated Power Domains:** The electrical architecture is split into two galvanically isolated domains: “Dirty” powers high-noise actuators and fans, “Clean” powers logic boards, sensors, and the central computer, preventing back-EMF interference.
- **The Arterial Network (Cabling):** A “Spinal” cabling topology runs a central trunk up the torso with branches to limbs. All cables are internally routed through hollow bones, using high-flexibility shielded cables that endure millions of flex cycles.

System Considerations

- **Regenerative Braking:** Motors act as generators during deceleration. The BMS handles reverse current safely, either dumping it or recharging the battery cells.

4.3 Brain (Compute Unit)

Functional Objective: The Brain is the centralized processing hub responsible for fusing sensory data, calculating kinematics, and generating motor commands in real-time. It prioritizes “Latency Determinism” to ensure critical calculations complete within microsecond windows, preventing the robot from falling.

Technical Design (Heterogeneous Computing)

- **Parallel Processing Architecture:** Workload is divided across specialized hardware accelerators.
- **GPU (Graphics Processing Unit):** Handles heavy parallel tasks like stereo camera processing, neural networks, and SLAM mapping.
- **CPU (Central Processing Unit):** Manages sequential tasks such as high-level planning, scheduling, and OS management.
- **Real-Time Microcontrollers (MCU):** Serve as the “Spinal Cord,” executing control loops (balance, motor commutation) at 1 kHz+, ensuring stability even if the CPU freezes.

System Considerations

- **Thermal Integration:** The compute unit is thermally coupled to the aluminum structural frame, acting as a passive heat sink and avoiding bulky fans.

4.4 Consciousness (Software & Control)

Functional Objective: “Consciousness” represents the software stack and control architecture, governing why the robot moves. It transforms raw sensor data into meaningful actions, enabling navigation in unstructured environments and natural human-robot interaction through language and gestures.

Technical Design (The AI Stack)

- **Perception Layer:** Converts raw data into semantic understanding, including occupancy grids and object classification.
- **Motion Planning (The “How”):** Solves inverse kinematics, using Model Predictive Control (MPC) to optimize walking gait and prevent collisions.
- **Cognitive Layer (The “Why”):** LLMs/VLMs process human commands, decomposing abstract goals into sequences of executable primitives.

System Considerations

- **Safety Watchdogs:** Independent software reflexes monitor tilt angles and collision forces, executing emergency stops or bracing maneuvers to protect hardware and bystanders.